



Macro Forecasting with the TFT and Fed Speeches

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Problem Setting & Motivation

- Macro Forecasting hard
- Forward Guidance / Superior Info
- Deep Learning Context

Data

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- Macro Data
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 - E.g.: CPI, UNRATE, GDP, Exchange Rates, etc.

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- Meta Data
 - Macro Metadata (Popularity, Frequency, Units)
 - FOMC Dissent
 - Days till next FOMC Meeting, etc.

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 - Days till next FOMC Meeting, etc.
- Embedded Fed Speeches
 - E.g. *“Remarks by Mr Roger W Ferguson Jr, Vice-Chairman of the Board of Governors of the US Federal Reserve System, before the National Economic Association in Boston on 7 January 2000.”*
 - Embedded (via FOMC-RoBERTa, FinBERT and Ollama,)
 - Dimensionality Reduction (None, PCA, Factor Analysis)
 - Speech Aggregation (mean, exp. decay, (context)attention-based)

Show Data...

by plotting, e.g.

Models

- Statistical Benchmarks (AR, ARIMA)
- Temporal Fusion Transformer (Lim et al., 2021)

Hyperparameter Tuning

show the grid, best values, etc.

Results

- Incl. Tables, Figures

Robustness Checks

- Shuffling Embeddings
- Using New unrelated Text (German Texts of Franz Kafka)

Future Work

- More Hyperparam Tuning
- Try out SSMs for Forecasting
- Add Vintages for Macro Data for higher Accuracy
- Scrape more speeches (to have a longer Train Horizon)

Bibliography

Lim, B., Arık, S. Ö., Loeff, N., & Pfister, T. (2021). Temporal Fusion Transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764. <https://doi.org/https://doi.org/10.1016/j.ijforecast.2021.03.012>